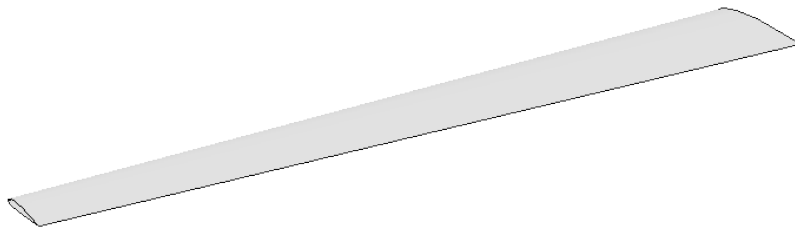


CHALMERS



Finite Element Method - Structures, TME245

Computer exercise 2: 3D shell analysis

Instructions

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Preface

Within this computer exercise, a 3D shell problem is considered. More in detail, you are expected to determine a suitable minimum *even* thickness of two versions of a shell wing design to meet a stress-based and an elastic buckling design criterion. In the oral report, you should be able to explain your considerations when it comes to boundary conditions, element type, element size, load application etc. No report is to be submitted. Please, however, note that you need to sign your group up for a presentation slot on 12 March. Do this here: <https://choodle.portal.chalmers.se/N48ZWDPuMeFed168>. Note that each 30 min slot can accommodate at most 8 groups.

The following file is made available to support this computer exercise:

`wing_geometry.cae`

Learning outcomes: The learning outcomes of this project are that you should get acquainted with how 3D shell analyses can be performed using a commercial software (in this case Abaqus CAE), and get more comfortable with identifying relevant boundary conditions for a "real" problem.

Göteborg in March 2026

Martin Fagerström

Problem definition

An university professor has set out to design his own airplane. His target for this small airplane is that its total mass should be at most 10,000 kg. As a starting point, they want to design the wings of the airplane, and they have found a CAD file with a wing design from <https://grabcad.com> that they want to use. The design is such that the wing is having a thinwalled hollow cross-section with even thickness. As a comparison, they also want to evaluate a completely flat design of the wing with the same planform (same shape if seen from above).

The professor has opted for a steel design, with the following parameters: $E = 210$ GPa, $\nu = 0.3$ and initial yield stress $\sigma_y = 500$ MPa.

A geometrical description of both wing designs to be considered are supplied in the Abaqus CAE file `wing_geometry.cae`. In the file, the wing midsurfaces have been designed in a CAD software using centimetre units.

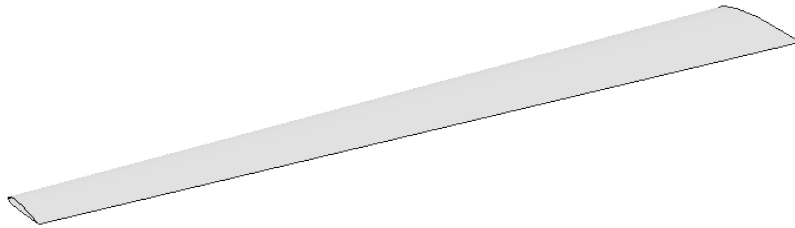


Figure 1: a) The hollow wing geometry to be considered in Computer Exercise 2.

From the given midsurface geometry, define the complete finite element model in Abaqus. To get the proper loads for each load case, make sure to solve the buckling load case *before* the static stress analysis. If you solve it the other way around, any loads defined in the static stress analysis step will be carried over as a constant load for the buckling analysis (which you don't want!)

So it is recommended that you solve the task using the following steps:

- Define the material behaviour.
- For each wing design, create a shell section with an initial thickness of your choice (to be changed) and assign that to the corresponding part.

- Create instances of the parts in the assembly module. Here, you can either choose to create two models, one for each part, or keep both parts as disconnected and with separate boundary conditions.
- Create a first buckling load case. This can be done in the step module via **Step -> Create -> Procedure Type -> Linear Perturbation -> Buckle**. The Lanczos eigensolver should work well for this problem. Please choose a suitable number of positive eigenvalues and eigenmodes to be analysed (>3).
- Create a second static load case (automatic time stepping over a quasi-time period from 0 to 1 should suffice since the load case is linear elastic).
- Define the appropriate boundary conditions separately for each step (the Linearised Prebuckling Analysis and the general static load case, respectively). Please note that even though the boundary conditions are the same for both cases, at least separate shell loads need to be defined for each step (however with the same magnitude and direction). Any displacement boundary conditions can be specified for the initial step and the propagated forward.

Hint: To compute the area of a separate section of the wing, one way to do it is to create a set defined by that geometrical section. Then, you can query mass properties for this section (or this set) via **Tools -> Query -> Mass properties** and then select the appropriate set.

- Discretise the domain in a suitable way using the shell element type of your choice. Make sure to motivate the type of discretisation (size and element type) during the oral reporting. When presenting the results, also make sure that this is done for a converged mesh.
- Note that useful Abaqus instructions can be found in the first recorded shell lecture from 2022, available on Canvas!

Task: Dimensioning of the necessary uniform wall thickness for each wing design using Abaqus

Use 3D shell analysis in Abaqus CAE to determine a minimum *even* thickness of the two different wing designs such the total lift force of each wing corresponds to half the total weight of the plane. It is your job to define the load in a suitable way, so please prepare your argumentation before the oral reporting! Please also use a safety factor against plastic yielding and elastic buckling of $n = 1.5$.

Here, you must consider, and later motivate and discuss, the following aspects:

- How did you apply the displacement boundary conditions? Why?
- How did you apply the load? What is the magnitude? Why?
- How do the results differ if you use shell elements with linear or quadratic displacement and rotation approximations? What mesh size is necessary in each of the two cases to have a reliable result for each?

Please note that for the oral presentation you need to prepare two result files that you can show, one where you have used shells with linear approximation and one where you have used shells with quadratic approximation.

- How did you determine the minimum even thickness? What results did you look at, and how did that result in the thickness you propose?
- Looking at the amount of displacements at the design load, do you notice any significant difference between the two designs if they both have the same mass (which approximately means that the flat case has twice the thickness of the other design)?